

PARTH SARTHI SAXENA

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PROFESSIONAL SUMMARY

Software engineering student with strong foundations in C++, Python, data structures & algorithms, and backend development. Experienced in building full-stack and data-driven applications with secure authentication, REST APIs, MongoDB integration, and modular system design. Built scalable backend workflows for travel matching and messaging platforms while solving 270+ LeetCode problems to strengthen problem-solving and algorithmic thinking. Interested in backend engineering, scalable systems, and developing analytical software solutions with strong engineering fundamentals..

EDUCATION

JECRC Foundation, Jaipur

May 2027

B.Tech – Electronics & Communication Engineering | CGPA: 8.21/10 (~3.4/4.0)

TRADING & QUANTITATIVE EXPERIENCE

Quantitative Systems Researcher | Self-Employed

2024 – Present

- Developed Python-based analytical systems for evaluating rule-based strategies using historical data, statistical metrics, and performance visualization workflows
- Built modular backtesting pipelines to simulate strategy behavior across different market conditions, including trade execution, drawdown analysis, and cumulative return tracking
- Applied probability-driven analysis and structured data evaluation to optimize decision-making workflows and assess performance consistency across large datasets
- Maintained structured datasets and reusable analysis utilities for historical simulation, parameter testing, and performance attribution

TECHNICAL PROJECTS

Roam Around -- Travel Collaboration & Exploration Platform

MERN

- Developed a full-stack travel matching platform enabling users to discover compatible travel partners for road trips, hikes, and motorcycle rides based on trip preferences and activity type
- Built secure authentication and session management using JWT, bcrypt password hashing, and httpOnly cookies with protected route handling
- Designed and integrated RESTful APIs for trip publishing, match management, and persistent real-time-style messaging workflows using Node.js and MongoDB
- Engineered MongoDB persistence architecture for users, trips, matches, and threaded chat messages with scalable collection design and state restoration across sessions
- Implemented modular backend workflows supporting trip publishing, match acceptance/rejection, and chat history retrieval with structured database interaction patterns
- Planned extensible matching infrastructure for future compatibility scoring using destination overlap, travel pace, budget, and behavioral preference analysis

Quantitative Strategy Backtesting Engine

Apr 2025 – May 2025

- Developed a Python-based backtesting engine to simulate and evaluate EMA-driven trading strategies on historical market data using pandas and numpy, with configurable trade execution and strategy parameters
- Built modular analytical workflows for calculating cumulative returns, drawdown, win rate, and risk-adjusted performance metrics, along with visualization of equity curves and strategy behavior across different market conditions
- Structured the framework to support extensible strategy integration and historical performance analysis through reusable data-processing and simulation pipelines

SKILLS

Programming: Python (pandas, numpy), C++, SQL, Node.js, MongoDB, Git, REST APIs, JWT Authentication, Tailwind CSS

Core Fundamentals: Data Structures & Algorithms, OOP, DBMS, Operating System

Concept: Problem Solving, API Integration, Data Analysis, Authentication Systems, Full-Stack Development

Interests: Software Engineering, Backend Systems, Quantitative Applications, Analytical Software Development